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Among all constants, the speed of light is the most famous. No matter how fast you go, you will never overtake a beam of light. Why not? No experiment ever conducted has ever revealed an object of any form reaching the speed of light. Well-tested laws of physics predict and account for that fact. I know these statements sound closed-minded. Some of the most bone-headed, science-based proclamations in the past have underestimated the ingenuity of inventors and engineers: "We will never fly." "Flying will never be commercially feasible." "We will never split the atom." "We will never break the sound barrier." "We will never go to the Moon." What they have in common is that no established law of physics stood in the their way.

The claim "We will never outrun a beam of light" is a qualitatively different prediction. It flows from basic, time-tested physical principles. Highway signs for interstellar travelers of the future will justifiably read:

> The Speed of Light:
> It's Not Just a Good Idea
> It's the Law.

Unlike getting caught speeding on Earth roads, the good thing about the laws of physics is that they require no law enforcement agencies to maintain them, although I did once own a geeky T-shirt that proclaimed, "OBEY GRAVITY."

All measurements suggest that the known fundamental constants, and the physical laws that reference them, are neither time-dependent nor location-dependent. They're truly constant and universal.

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Many natural phenomena manifest multiple physical laws operating at once. This fact often complicates the analysis and, in most cases, requires high-performance computing to calculate what's going on and to keep track of important parameters. When comet Shoemaker-Levy 9 plunged into Jupiter's gas-rich atmosphere in July 1994, and then exploded, the most accurate computer model combined the laws of fluid mechanics, thermodynamics, kinematics, and gravitation. Climate and weather represent other leading examples of complicated (and difficult-to-predict) phenomena. But the basic laws governing them are still at